

Endogenous Reachability Collapse under Stochastic Forcing (ERC-S):

Emergence of a Probabilistic Recovery Boundary in (AUC, Duration) Space

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AI-assisted modeling and analysis using ChatGPT and Grok

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Abstract

We extend a minimal dynamical model of recoverability by introducing stochastic perturbations to examine how a previously sharp recovery boundary changes under noise. While the deterministic system exhibits an abrupt transition, the addition of noise leads to the emergence of a probabilistic boundary in the (AUC, duration) space. Regions of high and low recovery probability remain distinguishable, separated by a transition zone where outcomes become variable. Recovery probability shows non-monotonic dependence on input duration for fixed total load, indicating that temporal structure plays a critical role. The recovered state remains present across all conditions, but may become dynamically inaccessible within the observation window. These results suggest that recoverability may depend on the interaction between input magnitude and temporal profile, rather than on total load alone.

1. Introduction

In minimal dynamical systems, recoverability is often interpreted as the ability of a system to return to a baseline state following perturbation.

In ERC-D, a sharp recoverability boundary was observed under deterministic conditions, separating regimes where trajectories return to baseline from those where they do not.

This work extends that framework by introducing stochastic perturbations, with the goal of examining whether the recoverability boundary persists and how its structure is modified under noise.

The aim is not to establish generality across physical or biological systems, but to provide a minimal demonstration that structured probabilistic recovery boundaries can emerge under simple dynamical constraints.

2. Model

We consider the same minimal dynamical system introduced in ERC-D, with the addition of stochastic forcing.

The system is described by a single state variable $v(t)$, evolving according to:

$$dv/dt = (1 - v)/\tau_r - E(t) + \sigma \xi(t)$$

where:

- $v(t)$ represents the system state
- τ_r is the intrinsic recovery timescale

- $E(t)$ is an external input (forcing)
- $\xi(t)$ is Gaussian white noise with zero mean and unit variance
- σ controls the noise amplitude

The input $E(t)$ is constructed such that the total delivered load (AUC) is fixed:

$$\text{AUC} = \int E(t) dt$$

while its temporal profile varies through changes in input duration and corresponding amplitude adjustments.

The system is evaluated under inputs with:

- fixed total magnitude (AUC)
- varying duration
- corresponding amplitude adjustments to preserve total load

Recovery is assessed across multiple stochastic realizations, allowing estimation of recovery probability for each (AUC, duration) configuration.

Recovery is defined operationally as:

$$v(t) \geq v_{\text{threshold}} \text{ for some } t \leq T_{\text{obs}}$$

where $v_{\text{threshold}}$ is a predefined recovery level and T_{obs} is the observation window.

Recovery probability is estimated as the fraction of stochastic realizations satisfying this condition.

Implementation details

The external input $E(t)$ is constructed as a Gaussian pulse with fixed area (AUC), where duration is controlled by the pulse width and amplitude is adjusted accordingly to preserve total load.

For each (AUC, duration) configuration, multiple stochastic realizations are simulated.

Recovery is evaluated using:

- collapse condition: $\min_t v(t) \leq v_{\text{collapse}}$
- return condition: $\exists t \leq T_{\text{obs}}$ such that $v(t) \geq v_{\text{threshold}}$

Recovery probability is defined as the fraction of realizations that satisfy the return condition within the observation window.

3. Results

3.1 Emergence of a probabilistic recovery boundary

When stochastic perturbations are introduced, the sharp deterministic transition observed in the minimal model broadens into a probabilistic boundary in the (AUC, duration) space.

Regions of high recovery probability (≈ 1) and low recovery probability (≈ 0) remain clearly distinguishable. However, an intermediate transition region emerges in which recovery probability takes values between 0 and 1.

This suggests that, in this minimal model, near the regime boundary small perturbations may influence trajectory outcomes, leading to variability in recovery.

3.2 Non-monotonic dependence on input duration

For fixed AUC values, recovery probability does not vary monotonically with input duration.

In particular:

- Short-duration, high-amplitude inputs may remain recoverable
- Intermediate durations may lead to loss of recovery
- Longer durations at the same total AUC may restore recoverability

This non-monotonic structure, as observed in this model, indicates that recovery is not determined solely by total input magnitude, but by its temporal distribution.

3.3 Dependence on joint (AUC, duration) configuration

The recovery landscape exhibits an inclined transition region in the (AUC, duration) plane.

This indicates that neither AUC nor duration alone is sufficient to predict system behavior. Instead, recoverability depends on their joint configuration.

This observation is consistent with the hypothesis that, in this model, loss of recovery may emerge when internal recovery dynamics cannot track the temporal structure of the imposed perturbation.

3.4 Interpretation as loss of dynamic accessibility

Across all simulations, the recovered state remains part of the modeled state space.

However, in regions of low recovery probability, trajectories fail to return to that state within the observation window.

This is consistent with an interpretation in which:

- failure does not require removal of states
- but may instead reflect a loss of dynamic accessibility

This is consistent with an extension of the deterministic result observed in ERC-D.

3.5 Robustness under stochastic perturbations

The persistence of a structured recovery boundary under stochastic forcing suggests that the observed behavior is unlikely to be a numerical artifact of deterministic dynamics.

Instead, the system maintains a separation between regimes of high and low recoverability, with a transition region reflecting sensitivity to perturbations near the boundary.

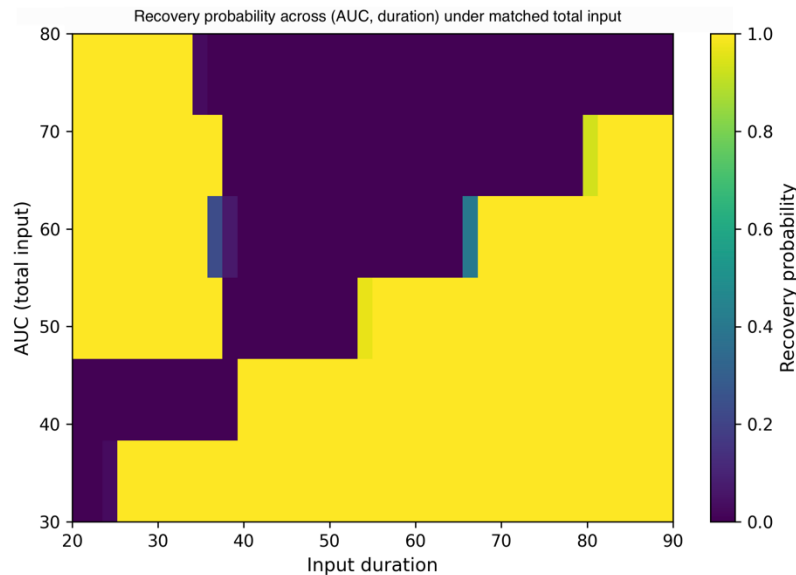


Figure 1. Recovery probability across (AUC, duration) under matched total input in a minimal dynamical system with stochastic perturbations.

Each point represents the probability of recovery estimated over multiple stochastic realizations for a fixed total input (AUC) and input duration. The input amplitude is adjusted such that total delivered load remains constant while its temporal profile varies.

Regions of high recovery probability (yellow) and low recovery probability (purple) are separated by a transition zone where recovery becomes probabilistic. The boundary is not aligned with either axis, indicating that recoverability depends on the joint configuration of input magnitude and temporal structure.

The recovered state remains part of the modeled state space across all conditions; however, in the low-probability region, trajectories fail to return to it within the observation window. This behavior is consistent with a loss of dynamic accessibility rather than removal of states.

The non-monotonic dependence on duration for fixed AUC suggests that intermediate temporal profiles may be more disruptive than both shorter and longer inputs of equal total magnitude.

4. Falsifiable Predictions

The model leads to the following testable predictions:

1. Persistence of structured boundary under noise
If stochastic perturbations completely eliminate structured differences in recovery probability, the model would not be supported.
2. Dependence on temporal structure
If recovery depends only on total input magnitude (AUC) and not on duration or temporal profile, the model would be challenged.
3. Non-monotonic response
If recovery probability varies monotonically with duration for fixed AUC across systems, the predicted structure would not hold.
4. Probabilistic transition near boundary
Near the regime boundary, small perturbations are expected to lead to variability in outcomes. Absence of such variability would challenge the framework.

5. Discussion

These results suggest that, in this minimal model, recoverability may be governed not only by the magnitude of perturbations, but also by their temporal structure and interaction with internal system dynamics.

The emergence of a probabilistic boundary under stochastic forcing indicates that the recoverability transition is not eliminated by noise, but instead becomes structured in terms of outcome likelihood.

The model remains minimal and does not assume a specific physical or biological mechanism. Further work is required to determine how these dynamics may manifest in real systems.

These results do not establish generality across systems, but provide a minimal demonstration that such behavior can emerge under simple dynamical constraints.

As in the deterministic case, the emergence of a structured boundary appears to depend on internal system dynamics rather than on external input alone.

These results are consistent with the hypothesis that recoverability depends on the temporal structure of perturbations relative to intrinsic recovery timescales, rather than on total input magnitude alone.

6. Conclusion

We show that, in a minimal dynamical system, a structured, probabilistic recovery boundary can emerge under stochastic perturbations. While the deterministic system displays a sharp transition, the addition of noise broadens this boundary into a region of variable outcomes.

This behavior is consistent with the possibility that recoverability may depend on the interaction between input magnitude and its temporal structure, and that failure may arise from loss of dynamic accessibility rather than removal of states.

This suggests that recoverability may be constrained by internal dynamical limits on resolving perturbations within finite time windows.

This work extends the deterministic framework introduced in ERC-D, where a sharp recoverability boundary emerges in a minimal system.